# Hyperbolic Hierarchy-Aware Prototype Network for On-Orbit Earth Surface Anomaly Detection From Single-Satellite Imagery

Fengcheng Ji, Kun Jia, Qiao Wang, Haishuo Wei, Zihang Jiang, and Can Xu

***Abstract*—Timely and accurate detection of Earth surface anomalies (ESAs) using single-temporal remote sensing data is critical for early warning and rapid emergency response. However, existing algorithms are designed mostly for specific ESA categories and suffer from ambiguous decision boundaries between anomalous and normal instances. Therefore, this study proposes a novel unified ESA detection (ESAD) model based on historical steady-state priors that quantify feature deviations as detection indicators, with enhanced discrimination achieved by amplifying the separation between anomalies and the normal Earth surface. First, an enhanced hyperbolic space network is developed, which can represent high-dimensional complex data in low-dimensional spaces by capturing hierarchical structures in images, thereby extracting richer features from remote sensing imagery. Then, these extracted features are aggregated under semantic guidance to construct the historical steady-state prior that characterizes normal Earth surface patterns, which is represented by a lightweight prototype set. Next, to further improve the representativeness of the steady-state prior, a semantic augmented contrastive clustering (SACC) module is introduced to refine the prototype set by reducing intraclass variability and increasing interclass separability. Finally, leveraging the geometric property of hyperbolic space which amplifies feature differences, a hyperbolic distance-based anomaly scoring function is proposed to quantify the anomaly level of test samples by measuring the hyperbolic distance between their feature representations and prototypes in prototype set. This enables accurate ESAD from single-temporal remote sensing image. The results indicate that the proposed method outperforms the currently popular methods in both accuracy and robustness and has potential for on-orbit deployment and applications. Code and dataset are available at https://github.com/Jifc1024/H2PNet**

***Index Terms*—Contrastive clustering, deep learning (DL), Earth surface anomaly detection (ESAD), hyperbolic space, on-orbit deployment.**

## I. Introduction

EARTH surface anomalies (ESAs) refer to Earth surface phenomena threatening the natural environment and social security, which disrupt the historical stable structure and feature distribution pattern over a certain period on the Earth's surface caused by the single or double influence of natural and human factors [1], [2], [3]. Over the past decades, the frequency of ESAs has risen markedly, which has led to severe consequences, including human casualties, property loss, and ecological damage [4], [5]. Therefore, rapid detection and response to these anomalies are critical for minimizing risks to human life, reducing economic losses, and mitigating long-term environmental effects [3], [6], [7]. Numerous studies have demonstrated that on-orbit processing can significantly enhance the responsiveness to ESAs by shortening the task execution chain [1], [8], [9], thus making on-orbit immediate ESA detection (ESAD) an important research direction. Unlike traditional ground-based processing methods, satellites have significantly lower storage capacity and computational efficiency. For example, the on-orbit storage capacities of Beijing-2, Sentinel-1, and Sentinel-2 are approximately 544 GB, 1410 GB, and 2 TB [10], respectively, which greatly limits the range of applicable algorithms, such as change detection (CD) methods that rely on historical data. Moreover, the limited computational efficiency also imposes greater demands on the efficiency of on-orbit models. Therefore, developing a lightweight ESAD method that is based on single-temporal remote sensing imagery is key to enabling on-orbit ESAD.

In recent years, advances in satellite technology, including broader observation coverage and more frequent revisits, have spurred a wide range of studies dedicated to ESAD based on single-temporal remote sensing imagery [11], [12], [13]. In the early stages, remote sensing index-based methods emerged as powerful tools for detecting ESAs because of their computational efficiency and strong interpretability. By analyzing the spectral response characteristics of anomalous events in remote sensing imagery, researchers have been able to develop targeted feature extraction and classification indices, such as the black and odorous water index (BOI) [14], the burned area index (BAI) [15], and the MODIS red tie index (MRI) [16], thereby enabling accurate detection of such phenomena from single-temporal remote sensing image. Nevertheless, remote sensing index-based methods typically depend on manually designed features and exhibit limited generalizability for scenes with complex backgrounds. To address these limitations, deep learning (DL)-based methods have emerged as powerful alternatives [17], [18], [19]. These studies focus on employing advanced techniques such as convolutional neural networks (CNNs) [20], [21], generative

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adversarial networks (GANs) [22], [23], and transformer networks [24], [25] to automatically capture the complex spatial, spectral, and semantic patterns of ESAs from remote sensing imagery, thereby improving detection accuracy and robustness. Among them, the design and development of models tailored to specific ESA types, such as forest fires [26], landslides [27], and plant diseases [28], represent important directions in current research. These methods often incorporate task-specific network architectures or supervision strategies to enhance the model's ability to identify and characterize the unique patterns associated with each ESA type. Given the high diversity of ESA types in real-world scenarios, various researchers have shifted their focus from single-type ESAD to joint detection models that are capable of handling multiple known ESA types [7], [29], [30]. They aim to leverage a unified network architecture to capture the semantic differences and interrelationships among various ESA events collaboratively, thereby enabling efficient identification and differentiation of multiple ESA types. By embedding prior knowledge of possible anomalies into the learning process, these models can enhance their sensitivity to similar anomaly events during deployment. However, this approach inherently limits the generalization ability of the models for unseen anomaly types that fall outside the predefined classes, thus increasing the risk of missed detections.

Background modeling-based methods are feasible approaches that enable the indirect detection of anomalous pixels by learning the statistical properties or structural patterns of background regions within the image, which offer a promising solution to the aforementioned challenges. This is primarily because background pixels in remote sensing imagery typically exhibit specific feature distributions, whereas anomalies often exhibit inconsistent patterns that deviate from these distributions [5]. On the basis of their background information extraction strategies, existing methods can be categorized into statistical modeling-based methods [31], [32], geometric modeling-based methods [33], [34], and DL-based methods [35], [36]. In statistical modeling-based methods, the primary assumption is that background information follows a specific statistical distribution, whereas anomalous objects deviate from this distribution, with Reed–Xiaoli (RX) algorithm-based methods being the most typical [37]. In contrast, geometric modeling-based methods typically assume that background information can be represented by typical spectral vectors or feature vectors, whereas anomalous objects cannot be adequately characterized [38]. For DL-based methods, they first model the discriminative feature distributions of the image background through the design of novel network architectures and then discriminate anomalous pixels via reconstruction error or uncertainty measures [39]. While background modeling-based anomaly detection methods can eliminate the reliance on anomaly types, existing methods generally focus only on spectral and spatial dimensional features while neglecting the steady-state characteristics of the background reflected in the temporal dimension. This results in the extracted anomalies being primarily "potential ESAs" that deviate from local patterns or the background, rather than true ESA events.

To achieve truly type-agnostic ESAD, only a few studies have recently shifted their focus toward constructing a prior of the normal Earth surface from historical data and detecting anomalous pixels that deviate from this prior. For example, Wei et al. [2] were the first to propose the idea of using the mean and standard deviation of historical image features, such as the NDVI and NDWI, as prior knowledge and detecting ESAs by quantifying the deviations of current image features from this prior. Furthermore, they proposed a method for constructing prior within a multidimensional feature space and employed the distances and directions of sample shifts in this space as criteria for ESAD [8]. However, these methods rely primarily on spectral features while overlooking spatial information, which limits the associated model's ability to capture local structural information, thus rendering it susceptible to interference from local noise or subtle perturbations. To extract more robust feature priors, Xu et al. [5] proposed directly leveraging the SAM model to extract deep semantic features from images, which showed promising performance. Nonetheless, the SAM model lacks domain-specific knowledge in remote sensing and exhibits relatively low computational efficiency under limited resources, which constrains further improvements in detection accuracy and poses challenges for on-orbit deployment. In addition, existing methods focus primarily on the construction of prior knowledge while overlooking the issue of ambiguous decision boundaries caused by the high feature similarity between anomalous and normal instances.

Therefore, a novel ESAD model (termed H2PNet) is developed in this study, which integrates historical steady-state prior construction of normal Earth surface with an enhanced feature separation strategy for rapid and accurate ESAD from single-satellite imagery. Specifically, during the training phase, H2PNet can leverage its hierarchical perception capability to learn richer feature representations of the normal Earth surface from historical data under the guidance of semantic information. These features are then encoded into a stable-state prior and stored in the prototype set. In addition, a semantic augmented contrastive clustering (SACC) module is introduced in this study to reduce the variability of intraclass features and enhance the separability of interclass features, thus making the prior more distinct and representative. During the inference phase, H2PNet can amplify feature differences and calculate anomaly scores (ASs) by measuring the minimum hyperbolic distances between the feature representation of test samples and the prototype set, thereby enabling immediate ESAD from single-temporal remote sensing data. On the one hand, H2PNet can reduce reliance on specific anomaly types and enhance adaptability to previously unseen anomaly events by modeling normal Earth surface patterns. On the other hand, by leveraging the exponential mapping of hyperbolic space, the model can significantly amplify the differences between distinct features, thereby effectively mitigating the effects of small discrepancies between anomalous and normal features on the accuracy of ESAD. In this way, the proposed method can offer a feasible solution for on-orbit ESAD under the resource-constrained conditions of satellite systems by

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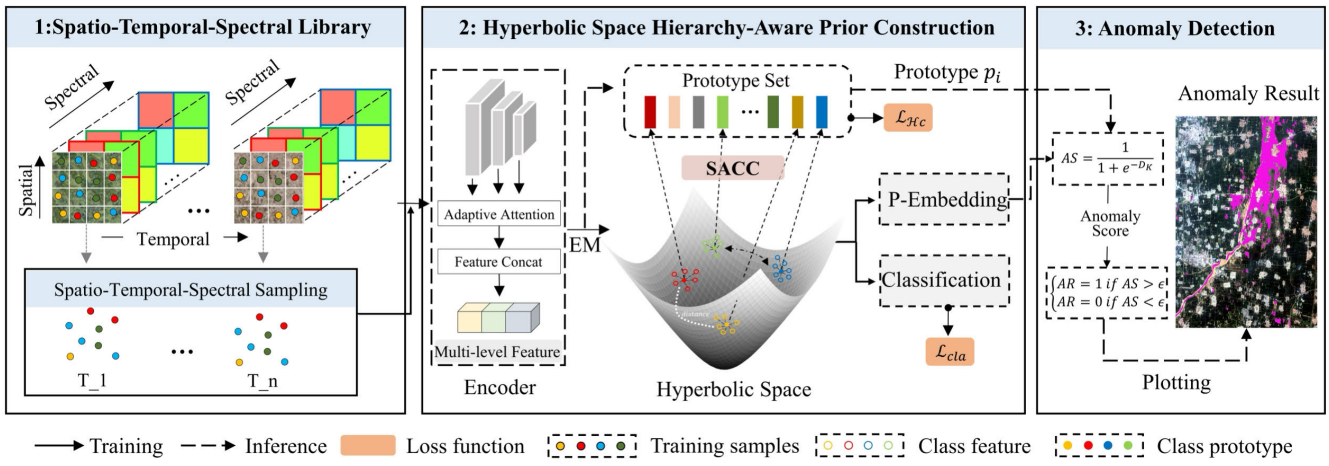



Fig. 1. Overview of the proposed H2PNet framework.

constructing a lightweight prior on the ground and deploying it for inference on the onboard platform.

## II. Methodology

The proposed framework comprises two main components: a prior construction module based on hyperbolic space hierarchical perception and an ESAD module that performs anomaly scoring via hyperbolic distance (see Fig. 1). The first module focuses on modeling the steady-state prior of the normal Earth surface using the proposed H2PNet and encodes them as a lightweight prototype set. The second module performs anomaly scoring by computing the minimum hyperbolic distance between the feature representations of test samples and each prototype in the prototype set, thereby enabling precise localization and identification of anomalous pixels.

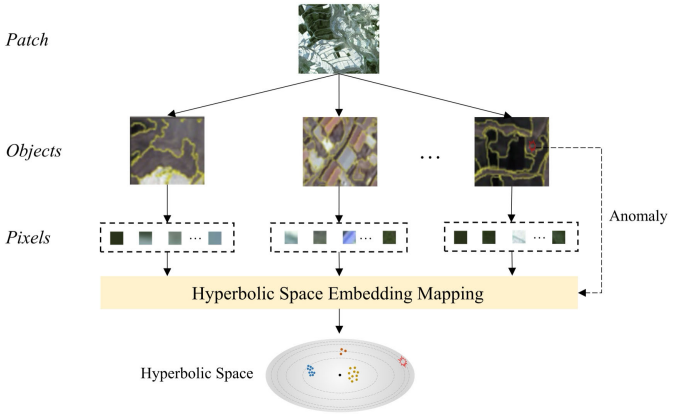



Fig. 2. Visual examples of hierarchical relationships in a remote sensing image.

### A. Preliminaries and Motivation

In traditional DL models, as the network parameters are iteratively optimized and updated, the model continuously learns new feature representations, whose embedding distributions in high-dimensional Euclidean space may form complex and extensive structures. However, the intrinsic geometric properties of Euclidean space (such as isotropy and flatness) impose limitations on its representational capacity in high dimensions. For instance, the volume of a hypersphere with radius $r$ grows polynomially with the dimension $d$ and is approximately proportional to $r^d$. This limited spatial scalability makes accommodating complex structures or large-scale feature distributions difficult, which often leads to interclass overlap and insufficient feature representation, thereby impairing the model's discriminative ability and generalization performance.

Unlike Euclidean space, hyperbolic space, which is characterized by constant negative curvature, exhibits exponential volume growth with the radius $r$ (e.g., $\sim e^{(d-1)r}$), thus offering a substantially broader embedding space in high dimensions [40]. This geometric property endows hyperbolic space with a superior ability to represent large-scale, high-dimensional features, which enhances the separability between classes and, thus, mitigates feature aliasing problems in Euclidean embeddings. For ESAD, anomalies typically exhibit feature differences compared with the normal Earth surface. The exponential expansion geometric property of hyperbolic space causes the same feature distance to be nonlinearly stretched after hyperbolic mapping, thereby amplifying these differences and facilitating the separation of anomalous events from the normal characteristics of the Earth's surface. In addition, the geometric property of exponential growth in hyperbolic space is similar to the growth of tree nodes. Thus, hyperbolic space is considered to have strong potential for exploring hierarchical structures [41]. This enables hyperbolic space-based models to preserve the topological structure and semantic hierarchy of data without requiring high-dimensional embeddings, which has been widely confirmed in previous studies [42], [43]. In the field of remote sensing, the three distinct image processing units of pixel–object–patch naturally possess such hierarchical characteristics. Effectively capturing this hierarchical structure is crucial for enhancing the model's understanding of the spectral, spatial, and semantic relationships of land cover (LC), modeling normal Earth surface features, and ultimately detecting anomalous events, as shown in Fig. 2.

In summary, the application of hyperbolic space to ESAD offers two main advantages: 1) the suitability of

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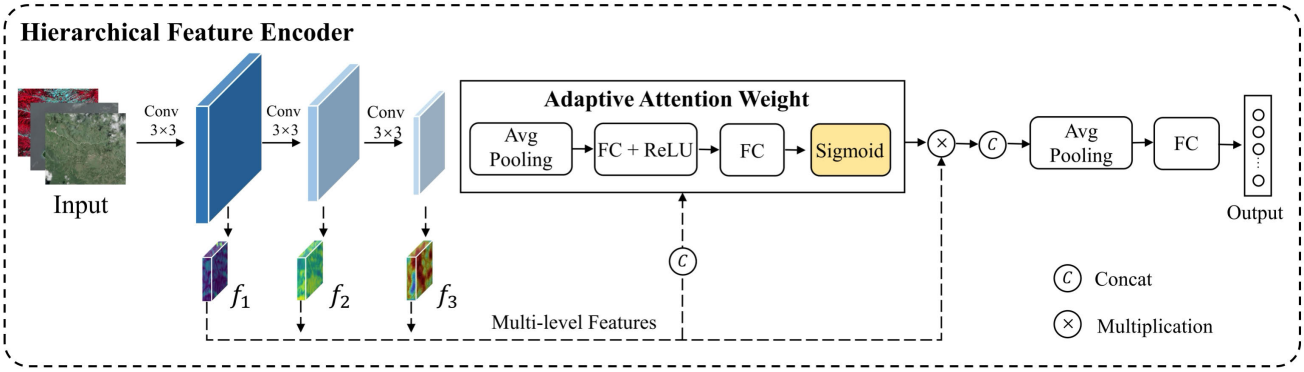



Fig. 3. Workflow of the HFE module.

hyperbolic space for modeling hierarchical structures facilitates the extraction of richer feature representations of the normal Earth surface from remote sensing imagery, thus offering structural support for anomaly discrimination and 2) the geometric property of exponential growth in hyperbolic space can amplify feature differences between normal and anomalous samples through nonlinear stretching, thereby mitigating the effects of ambiguous decision boundaries on ESAD performance.

### *B. Enhanced Hierarchical Feature Embedding-Based Hyperbolic Space Network*

As shown in Fig. 1, the feature extraction module of H2PNet comprises two main parts: an encoder and a hyperbolic space hidden representation. To further enhance the model's ability to extract hierarchical features, in this study, a hierarchical feature encoder (HFE) is developed to extract multilevel features explicitly from images, which are then mapped into hyperbolic space (see Fig. 3). Specifically, the encoder consists of three sequential convolutional layers, each followed by an ReLU activation function, to extract features $f_i$ at various levels

$$f_i = \mathrm{ReLU}\left(\mathrm{Conv}_i\left(f_{i-1}\right)\right), \quad i = 1, 2, 3. \tag{1}$$

Subsequently, an adaptive attention mechanism is introduced to assign distinct weights $w_i$ to features from these levels

$$\hat{f}_i = w_i \cdot f_i, i = 1, 2, 3 \tag{2}$$

where

$$w_i = \mathrm{Sigmoid}\left(\mathrm{FC}_2\left(\mathrm{ReLu}\left(\mathrm{FC}_1\left(\mathrm{AvgPool}\left(f_i\right)\right)\right)\right)\right) \tag{3}$$

where Sigmoid$(\cdot)$ represents the sigmoid activation function, $\mathrm{FC}_1$ and $\mathrm{FC}_2$ represent two fully connected layers, and AvgPool$(\cdot)$ denotes average pooling. These weighted features $\hat{f}_i$ are then integrated using a Concat operation to achieve effective multilevel feature fusion

$$f_c = \mathrm{Concat}\left(\hat{f}_1, \hat{f}_2, \hat{f}_3\right). \tag{4}$$

Finally, the fused features are processed through a pooling layer and a fully connected layer to obtain feature embeddings $F_k^{\mathbb{E}}$ in the Euclidean space

$$F_k^{\mathbb{E}} = \mathrm{FC}\left(\mathrm{AvgPool}\left(f_c\right)\right). \tag{5}$$

To optimize the proposed H2PNet in hyperbolic space, finding a mapping function from Euclidean space $\mathbb{E}$ to hyperbolic space $\mathcal{H}$ is necessary. Specifically, in this study, the Poincaré ball (a commonly used representation of hyperbolic space with excellent geometric properties) model with negative curvature $c$ is adopted as a representative model of hyperbolic space. To map from Euclidean space $\mathbb{E}$ to hyperbolic space $\mathcal{H}$, an exponential map (EM) is typically adopted. Given a Euclidean feature embedding $F_k^{\mathbb{E}}$, the closed-form expression of the EM is defined as

$$\begin{aligned} F_k^{\mathcal{H}} &= \exp_h^c\left(F_k^{\mathbb{E}}\right) \\ &= h \bigoplus_c \left(\tanh\left(\sqrt{c}\frac{\lambda_h^c \|F_k^{\mathbb{E}}\|}{2}\right)\frac{F_k^{\mathbb{E}}}{\sqrt{c}\|F_k^{\mathbb{E}}\|}\right) \end{aligned} \tag{6}$$

where $F_k^{\mathcal{H}}$ denotes feature embedding in hyperbolic space and $h$ denotes a fixed base point in hyperbolic space that can be simply set to $\mathbf{0}$, as in [44]; then, $\lambda_h^c = 2/(1 - c\|h\|^2)$ is equal to 2. $\|\cdot\|$ denotes the norm, and $\bigoplus_c$ represents the addition operation. However, unlike Euclidean space, hyperbolic space is not a vector space. That is, addition, multiplication, and other operations defined in Euclidean space cannot be directly applied in hyperbolic space. Therefore, addition in hyperbolic space can be performed using Möbius addition. The Möbius addition $\bigoplus_c$ of two points $x$ and $y$ is defined as

$$x \bigoplus_c y = \frac{\left(1 + 2c\langle x, y\rangle + c\|y\|^2\right)x + \left(1 - c\|x\|^2\right)y}{\left(1 + 2c\langle x, y\rangle + c^2\|x\|^2\|y\|^2\right)} \tag{7}$$

where $\langle\cdot\rangle$ denotes the inner product. In this way, the mapping from Euclidean space $\mathbb{E}$ to hyperbolic space $\mathcal{H}$ is completed.

Like training strategies based on Euclidean space, the model parameters are optimized by iteratively learning from training samples under the guidance of semantic information on the Earth's surface. To guide the optimization process effectively, cross-entropy loss is adopted as the objective function

$$\mathcal{L}_{\mathrm{cla}} = -\frac{1}{K}\sum_{i\in|N^s|}\sum_{k\in|K|} y_i^{(k)} \log\left(\mathcal{H}_\theta\left(x\right)\right) \tag{8}$$

where $K$ denotes the number of semantic classes, $|N^s|$ represents the number of samples, $y$ represents the class label, and $\mathcal{H}_\theta(x)$ denotes the output of the hyperbolic space, where the values correspond to the hyperbolic distances from sample feature to each class prototype instead of traditional logits. It

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is worth noting that the distance between two points $x$ and $y$ in hyperbolic space is given by

$$D(x,y) = \frac{2}{\sqrt{c}} \operatorname{arc\,tanh}\left(\sqrt{c}\left\| -x \bigoplus_c y \right\|\right). \tag{9}$$

Equations (6)–(9) demonstrate that when two features in Euclidean space are mapped to hyperbolic space, the distance between them in hyperbolic space is notably amplified because of the introduction of nonlinear transformations, thereby enhancing the ability to express feature differences. For example, given three Euclidean feature vectors $F_1^{\mathbb{E}} = (0.5, 0.2)$, $F_2^{\mathbb{E}} = (2.0, 0.7)$, and $F_3^{\mathbb{E}} = (5.0, 2.0)$, their corresponding representations in hyperbolic space with curvature $c = -0.5$ obtained via (6) are $F_1^{\mathcal{H}} = (0.48, 0.19)$, $F_2^{\mathcal{H}} = (1.20, 0.42)$, and $F_3^{\mathcal{H}} = (1.31, 0.52)$, respectively. Although the Euclidean distances between $(F_1^{\mathbb{E}}, F_2^{\mathbb{E}})$ and $(F_1^{\mathbb{E}}, F_3^{\mathbb{E}})$ are relatively small (1.58 and 4.85), the corresponding hyperbolic distances are notably enlarged to 3.94 and 26.6, respectively. This nonlinear amplification effect intensifies as features move away from the origin, allowing hyperbolic space to better preserve hierarchical structures and improve interclass separability.

### C. Prior Construction Based on SACC

During the model optimization process, owing to the model's ability to extract class features, features of the same class gradually cluster in the hyperbolic space under the guidance of class semantics, whereas those of different classes progressively diverge. To construct the steady-state prior that characterizes the normal Earth surface patterns, this study first identifies structurally stable and intraclass compact feature clusters in the hyperbolic feature space, where each cluster represents a distinct class. The steady-state prior is then represented by the mean feature embedding of each class cluster and stored in the prototype set. Specifically, the prototype of class $k \in K$ in the prototype set can be constructed by computing the mean of feature embeddings, as expressed as follows:

$$p_k = \frac{1}{N^k} \sum_{i \in |N^k|} \mathcal{R}_k^i \tag{10}$$

where $\mathcal{R}_k$ represents the feature queue that stores the updated hyperbolic space feature embeddings, with $\mathcal{R}_k = \{F_{k,1}^{\mathcal{H}}, F_{k,2}^{\mathcal{H}}, \ldots, F_{k,N^k}^{\mathcal{H}}\}$, $\mathcal{H}$ represents hyperbolic space, and $N^k$ denotes the size of the feature queue of class $k$.

However, in hyperbolic space, the mean computation cannot be conducted directly by averaging the inputs, as this may lead to a result being outside of the manifold. According to [45], the hyperbolic mean can be computed from the *Einstein midpoint*, which has the most concise form in the coordinates of the Klein model. Therefore, by projecting the Poincaré ball model onto the Klein model, the *Einstein midpoint* can be easily computed, thus yielding the hyperbolic mean. This method is feasible because the geometric models of hyperbolic space are isometric and can be transferred between each other by mapping functions [42]. Specifically, the Einstein midpoint (hyperbolic mean) of $N^k$ samples of class $k$ is defined as

$$\mu_{klein}^k = \frac{\sum_{i \in N^k} \gamma_i x_i}{\sum_{i \in N^k} \gamma_i} \tag{11}$$

where $x_i$ denotes the feature embedding in the Klein model and $\gamma_i = (1 - c\|x_i\|^2)^{-1/2}$ represents the Lorentz factor. Moreover, the transition between the Poincaré ball model $x_{\mathbb{P}}$ and the Klein model $x_{\mathbb{K}}$ can be derived as $x_{\mathbb{K}} = 2x_{\mathbb{P}}/(1 + c\|x_{\mathbb{P}}\|^2)$. Therefore, based on (11), the prototype of class $k$ in the Poincaré ball model can be represented as

$$p_k = \frac{\mu_{klein}^k}{1 + \sqrt{1 - c\|\mu_{klein}^k\|^2}}. \tag{12}$$

The obtained prototype $p_k$ is subsequently collected to form a prototype set $\mathcal{P} = \{p_1, p_2, \ldots, p_k\}$. To make the obtained prototype set $\mathcal{P}$ more representative and distinguishable, an SACC module is designed, as shown in Fig. 4. First, a contrastive clustering loss $L_{\mathcal{H}c}$ is introduced to further enhance the compactness within classes and separability between classes in hyperbolic space

$$\mathcal{L}_{\mathcal{H}c}\left(F_k^{\mathcal{H}}\right) = \sum_{i \in K} \mathcal{H}\left(F_k^{\mathcal{H}}, p_i\right) \tag{13}$$

where

$$\mathcal{H}\left(F_k^{\mathcal{H}}, p_i\right) = \begin{cases} D\left(F_k^{\mathcal{H}}, p_i\right), & if\ i = k \\ \max\left(0, \Delta - D\left(F_k^{\mathcal{H}}, p_i\right)\right), & if\ i \neq k \end{cases} \tag{14}$$

where $\Delta$ denotes the distance threshold. In this way, when a sample shares the same class as the prototype $p_i$, the model minimizes the distance between them. When the class of the sample is inconsistent with that of the prototype in the prototype set $\mathcal{P}$ and its distance to a mismatched prototype in the prototype set $\mathcal{P}$ is less than a threshold $\Delta$, the model increases the distance between the two. Conversely, when the distance between the two is sufficiently large, the model no longer needs to be updated, and the loss term is set to $\mathcal{H}(F_k^{\mathcal{H}}, p_i) = 0$. Second, a semantic-enhanced prototype set dynamic updating strategy is developed. Specifically, the feature queue $\mathcal{R}_k$ for each class $k \in K$ is dynamically updated using a rolling replacement strategy. When this feature queue, which has capacity $N^k$, receives new features, the oldest stored features are removed to maintain the capacity limit. When all the features in the queue have been replaced by new features once, the corresponding prototype is updated to reflect the feature distribution of class $k$ of the current stage while reducing the effects of batch bias on prototype updates. In addition, to prevent drastic changes in the class prototype and enable gradual evolution, this study adopts a momentum-based update strategy

$$\mathcal{P}_{\text{update}} = \alpha \bigotimes_c \mathcal{P} \bigoplus_c (1 - \alpha) \bigotimes_c \mathcal{P}_{\text{update}} \tag{15}$$

where $\mathcal{P}_{\text{update}}$ is the prototype set updated according to the updated feature queue $\mathcal{R}_k$. $\alpha$ denotes the momentum parameter and is set to 0.9, and $\bigotimes_c$ represents *Möbius scalar multiplication*, which is defined as

$$r \bigotimes_c x = \frac{\tanh\left(r \cdot \tanh^{-1}\sqrt{c}\|x\|\right) \cdot x}{\sqrt{c}\|x\|} \tag{16}$$

where $r$ denotes a scalar factor and $x$ is a feature embedding in hyperbolic space. In this way, through the combination of the feature queue and the momentum mechanism, the prototype

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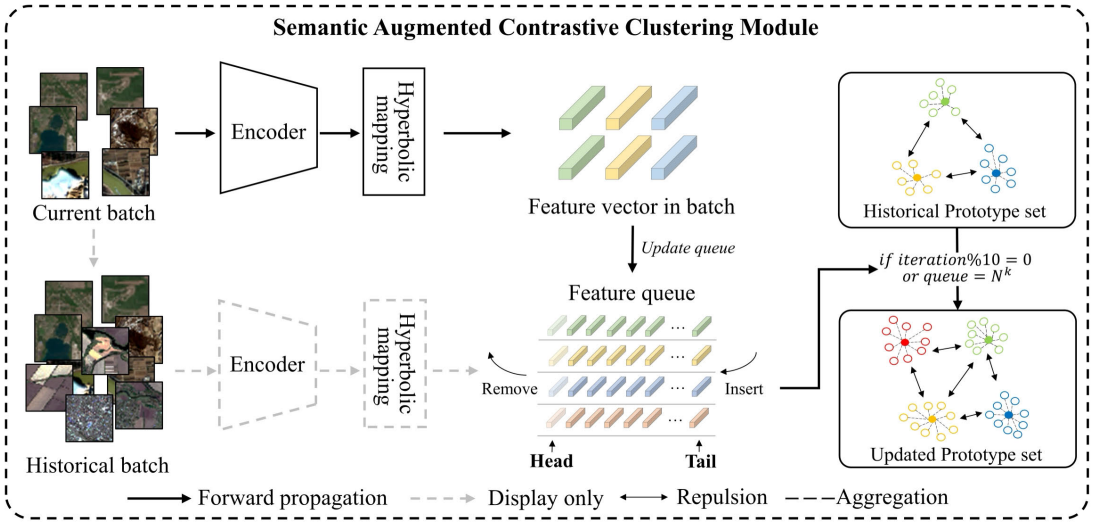



Fig. 4. Workflow of the SACC module.

set can capture both short-term local semantics and long-term global semantics, thereby providing enhanced semantic support for contrastive clustering loss. Contrastive clustering loss can also optimize the feature distribution to improve the representativeness of the prototype set, thus forming a mutual gain relationship closed loop. In addition, an extra condition is established for prototype updates: the prototype set is updated every ten iterations to ensure that it can respond promptly to feature changes.

Finally, the loss of H2PNet consists of two parts, namely, hyperbolic space classification loss $\mathcal{L}_{\text{cla}}$ and contrastive clustering loss $\mathcal{L}_{\mathcal{H}c}$, and is defined as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{\text{cla}} + \beta \mathcal{L}_{\mathcal{H}c} \tag{17}$$

where $\beta$ is the adjustment coefficient.

### D. Hyperbolic Distance-Aware ESAD

As previously discussed, the SACC module can effectively guide the model to learn feature representations with strong compactness and separability. The inherent hierarchical modeling capacity of hyperbolic space further enhances this property. Accordingly, the resulting prototype set $\mathcal{P}$, constructed from such discriminative features, can serve as a prior representation of the normal Earth surface characteristics. By leveraging this feature prior, ESA-related phenomena can be identified by measuring the degree to which feature representation of test samples deviates from the prototypes in the constructed prototype set. Specifically, the exponentially expanding geometric property in hyperbolic space can effectively amplify the distinction between anomalous and normal samples. Therefore, this study adopts the minimum hyperbolic distances between the hyperbolic embeddings of test samples and the prototypes in the prototype set $\mathcal{P}$ as a scoring criterion to detect ESAs.

First, the test sample is passed through H2PNet to obtain its feature embedding $F_k^{\mathcal{H}}$ in hyperbolic space. The hyperbolic distance between $F_k^{\mathcal{H}}$ and each $p_k$ in the prototype set $\mathcal{P}$ is subsequently computed using (9) and the results from a distance set $\mathcal{D} = \{D_1, D_2, \ldots, D_K\}$. These distances are then normalized to the (0.5, 1) range using (18), and the minimum normalized value is selected as the final AS

$$\text{AS} = \frac{1}{1 + e^{-D_k}}. \tag{18}$$

Then, by comparing the AS with the predefined anomaly threshold $\epsilon$, the location and scope of the final ESAs are determined, as expressed as follows:

$$\text{Anomaly} = \begin{cases} 1, & if\ \text{AS} > \epsilon \\ 0, & if\ \text{AS} < \epsilon. \end{cases} \tag{19}$$

Finally, to mitigate the effects of noise on the ESAD results, a $5 \times 5$ median filter is employed during postprocessing to improve the reliability of the results.

## III. Experimental Configuration

### A. ESAD Dataset

To support experimental validation, an ESAD dataset with eight typical ESA types was constructed. Detailed information about the ESAD dataset is provided in Table I, where "Area" refers to the area of the study region and "ESA-P" denotes the ratio of the anomalous area to the total image area.

To construct a reliable steady-state prior of normal Earth surface features, ground samples from three available Sentinel-2 images acquired within two months before the occurrence of the anomalous event (or two images if fewer were available) were selected. This strategy was intended to ensure that the prior features reflected the historical steady-state conditions of the Earth's surface while minimizing the influence of transient disturbances on ESAD. Specifically, Sentinel-2 images captured during the two months that preceded the anomalous event were first retrieved, and the three most recent available images were selected as reference data. In order to simulate conditions closer to the real on-orbit environment, characterized by limited computational resources and processing time windows, all cases in the ESAD dataset were based on top-of-atmosphere (TOA) reflectance data, which undergo only radiometric calibration and geometric correction, without

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TABLE I
DETAILED INFORMATION ON THE EXPERIMENTAL CASES IN THE ESAD DATASET

| ID | Anomaly Information | | | Data Description | | | |
|---|---|---|---|---|---|---|---|
| | Event | Location | Coordinates | Area | ESA-P | Occurrence time | Source |
| 1 | Landslide | Enga, PNG | -5.36, 143.44 | 471.17 $km^2$ | 0.02% | May 24, 2024 | Sentinel-2 |
| 2 | Forest fire | Shanxi, CHN | 35.90, 112.40 | 1861.57 $km^2$ | 1.30% | June 12, 2024 | Sentinel-2 |
| 3 | War | Krasnodar, RUS | 45.88, 40.12 | 967.21 $km^2$ | 0.69% | September 21, 2024 | Sentinel-2 |
| 4 | Green tide | Shandong, CHN | 36.02, 120.31 | 84.19 $km^2$ | 2.01% | July 13, 2019 | Sentinel-2 |
| 5 | Explosion | Illinois, USA | 42.39, -88.89 | 172.58 $km^2$ | 0.05% | June 14, 2024 | Sentinel-2 |
| 6 | Volcano | Rakata, IDN | -6.10, 105.43 | 220.10 $km^2$ | 0.83% | February 3, 2022 | Sentinel-2 |
| 7 | Flood | Hebei, CHN | 37.17, 114.66 | 593.07 $km^2$ | 9.27% | July 27, 2023 | Landsat-8 |
| 8 | Grassland fire | Sukhbaatar, MNG | 46.80, 118.89 | 832.01 $km^2$ | 33.47% | April 18, 2022 | Sentinel-2 |

TABLE II
SAMPLE DISTRIBUTION AND DATA ACQUISITION IN THE ESAD DATASET

| Category | Training Samples | | | | | Data Acquisition | |
|---|---|---|---|---|---|---|---|
| | Veg | Water | Built | Bare | Total | Normal Images | Anomaly Image |
| Landslide | 2054 | 773 | 634 | 0 | 3461 | April 8, 23, and May 23, 2024 | May 28, 2024 |
| Forest fire | 2243 | 657 | 903 | 0 | 3803 | April 24, May 14 and 24, 2024 | June 23, 2024 |
| War | 2076 | 713 | 844 | 0 | 3633 | September 8, 11, and 16, 2024 | September 21, 2024 |
| Green tide | 513 | 1129 | 618 | 326 | 2586 | June 11, 18, and 23, 2019 | July 13, 2019 |
| Explosion | 1529 | 613 | 527 | 415 | 3084 | April 30, May 10, and June 04, 2021 | June 14, 2021 |
| Volcano | 603 | 1050 | 0 | 375 | 2028 | December 28, 2021, and January 20, 2022 | February 3, 2022 |
| Flood | 924 | 485 | 816 | 279 | 2504 | July 18, and 26, 2023 | August 3, 2023 |
| Grassland fire | 1336 | 477 | 452 | 0 | 2265 | April 2 and 4, 2022 | April 19, 2022 |

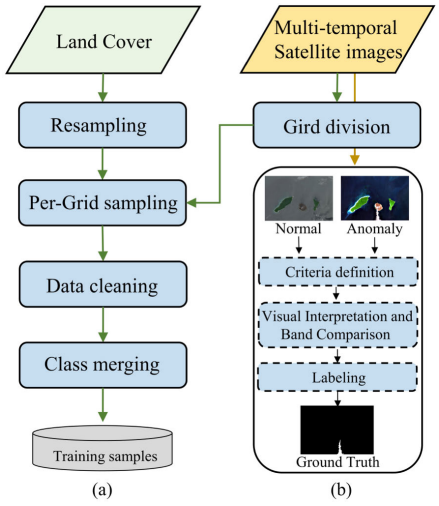



Fig. 5. Flowchart of dataset construction. (a) Training sample generation for steady-state prior construction. (b) Anomaly labeling process.

atmospheric correction. This configuration could substantially reduce the computational burden and processing costs associated with on-orbit atmospheric correction, thereby enhancing rapid response capability for anomalous event detection. Second, on the basis of the LC product released by Esri, the study area was divided into a 10×10 grid. Within each grid cell, ten samples were randomly selected for each LC type to mitigate the effects of spatial sampling imbalance on the construction of the prior. Considering the potential errors in the LC product, visual interpretation was further performed to manually refine the sample selection. For each LC type present within a grid cell, this study retained one to five samples based on interpretation results (if a grid cell was covered by clouds, no samples were retained from that cell). Then, the LC types were merged into four major categories: vegetation (including trees, grass, flooded vegetation, crops, and shrub), water (including water and snow/ice), built, and bare. Finally, all samples selected from the reference images were merged to construct the training samples, as shown in Table II. To alleviate the effect of class imbalance on model performance, data augmentation techniques such as mirroring and rotation were applied to minority classes.

For accurate labeling of anomalous events, this study developed a detailed anomaly labeling protocol, including data acquisition, labeling criteria definition, and labeling procedures [see Fig. 5(b)]. First, approximate anomaly locations were identified through news reports, followed by the acquisition of multitemporal remote sensing images covering preevent

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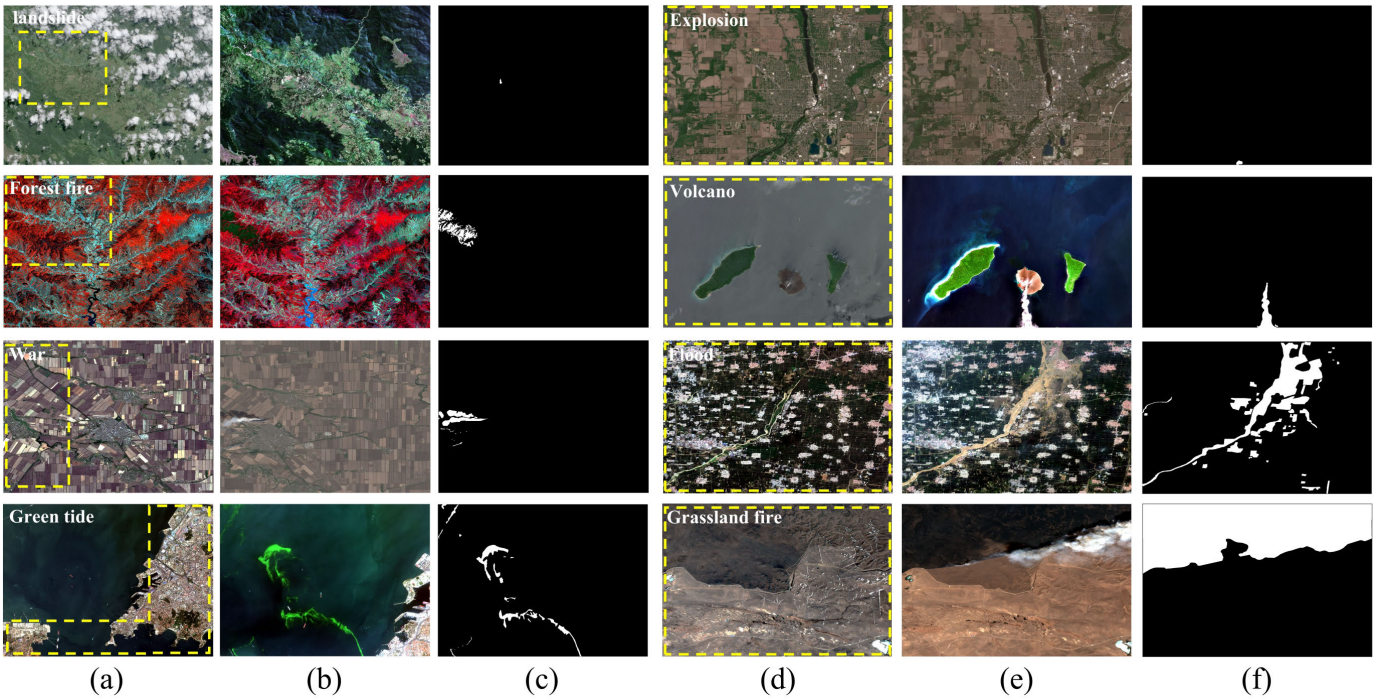



Fig. 6. Samples from the ESAD dataset. (a) and (d) Normal images before ESA occurrence. (b) and (e) Anomalous images. (c) and (f) Ground truth. The yellow dashed box represents the region used for constructing the historical steady-state prior.

and postevent periods (as detailed in Table II). All images underwent radiometric calibration and geometric correction. Second, type-specific labeling criteria were developed for each anomaly category in ESAD dataset: 1) for fire events, areas that were burning, had already burned, or showed visible smoke were marked as anomalies; 2) for explosion and war events, regions showing significant structural damage to buildings, as well as smoke and its associated effects, were marked as anomalies; 3) in the case of landslide, areas demonstrating evident surface displacement or showing exposed soil were identified as anomalies; 4) in the context of the green tide, large-scale algal blooms on the sea surface were designated as anomalies; 5) for volcanic activity, the anomalous areas were identified as those with volcanic plumes and lava flows; and 6) flood was marked in an area where the Earth surface was covered by turbid water. Following the establishment of labeling criteria, preliminary anomaly boundaries were delineated through visual interpretation of preevent and postevent images using various band combinations (e.g., true color and false-color infrared). Finally, a pixel-by-pixel band comparison was conducted for uncertain regions to determine the final anomalous areas. The labeling results are shown in Fig. 6.

Furthermore, remote sensing images are often affected by cloud cover, resulting in spatially incomplete historical observations for target areas and, thus, making it difficult to construct a steady-state prior directly across the entire detection area. To validate the applicability and flexibility of the proposed method under such practical constraints, four cases (landslide, forest fire, war, and green tide) in the ESAD dataset were designed with spatial inconsistency, where the region used for steady-state prior construction does not completely overlap with the detection regions. Specifically, the steady-state prior was constructed from local subregions within the detection area for landslide and forest fire but from adjacent surrounding areas for war and green tide. The other cases (explosion, volcano, flood, and grassland fire) in the ESAD dataset remained spatially consistent between steady-state prior construction and detection regions.

### B. Experimental Settings

*1) Comparison Methods:* To comprehensively verify the advantages of the proposed H2PNet, six anomaly detection methods, including two traditional methods and four DL-based methods, were selected. Specifically, the traditional methods are the RX-based method [37] and the Euclidean baseline (EBL)-based CD method [46]. Among them, the RX-based method is the most representative algorithm for anomaly detection based on single-temporal image. It assumes that the background spectrum follows a multivariate Gaussian distribution. It identifies anomalies by computing the mean vector and covariance matrix of the background and measuring the Mahalanobis distances between test pixels and the background's statistical profile, thereby quantifying the spectral deviation from the expected distribution. The traditional CD method was selected because it was the most commonly used means for detecting surface changes on Earth. Moreover, evaluating their performance can also facilitate understanding of the improvement caused by the high-level semantic features of DL. Furthermore, the DL-based methods include three advanced anomaly detection methods that are based on single-temporal remote sensing image, namely, TAEF [47], DFAN [48], DWT-P-CSA [49], and a DL-based CD method, CAAE [50]. The experimental results of all the comparison methods were obtained using their source codes. To ensure fairness

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and optimal performance in the context of this study, the implementations were fine-tuned, and their parameters were carefully adjusted. In addition, a $5 \times 5$ median filter was applied to the results of all methods to ensure the fairness of the comparison. It is worth noting that the CD-based method was adopted for a comparative experiment mainly to provide a more intuitive comparison of the advantages and disadvantages of the ESAD method, which is based on a single-temporal remote sensing image.

*2) Implementation Details:* For the hyperparameters of the proposed H2PNet, the initial learning rate was set to 0.001, and the Adam optimizer was used to adaptively adjust the learning rate during training. The maximum number of training epochs and the batch size were set to 300 and 32, respectively. In addition, the patch size of the model input was set to $5\times5$, the negative curvature $c$ of the hyperbolic space was 0.5, the distance threshold $\Delta$ was 0.5, and the adjustment coefficient $\beta$ was set to 0.1. Moreover, the training process was completed on a computer with an 11th-generation Intel[1] Core[2] i9-11900K processor and 128 GB of RAM, and the DL-based method was accelerated by using a 3060 GPU with 12 GB of memory. The proposed H2PNet employs PyTorch, and the version of Python was 3.7.

*3) Evaluation Metrics:* To compare the ESAD performance of the proposed H2PNet with those of the benchmark methods, this study employed various widely utilized metrics, such as precision, recall, and the $F1$ score, to assess the detection performance. Specifically, precision measures the proportion of samples predicted as positive by the model, which are truly positive, whereas recall measures the proportion of actual positive samples that are correctly identified by the model. The formulas for precision and recall are given as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{21}$$

where TP represents the number of correctly predicted anomalies, FP denotes the number of incorrectly predicted anomalies, and FN denotes the number of incorrectly predicted normal pixels. The $F1$ score is the harmonic mean of precision and recall, and is used to comprehensively evaluate a model's performance. The detailed formula for the $F1$ score can be expressed as

$$F1 = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}}. \tag{22}$$

## IV. Results and Analysis

### A. Ablation Experiment

To evaluate the contribution of key components in the proposed H2PNet method, ablation studies were performed by selectively removing or substituting individual modules and examining their effects on ESAD. The proposed method has three components: the HFE, the hyperbolic space-based feature embedding (HSFE) module, and the SACC module. For a fair comparison, the three parts were replaced with baseline alternatives while keeping other hyperparameters and training settings unchanged. Specifically, in the variant without HFE, the HFE was replaced with a single-level encoder that used only the final convolutional layer's output as input to the hyperbolic space. In the variant without HSFE, the hyperbolic space was replaced with Euclidean space to evaluate the contribution of the hyperbolic geometry to ESAD and feature representation. In addition, in the variant without the SACC module, prototype construction was designed to only use the mean of sample features, thereby evaluating the role of SACC in improving prototype representativeness and discriminability. All experiments were conducted on the ESAD dataset, and performance was evaluated using $F1$ metric.

Table III presents the performance improvements in ESAD by the proposed modules. As shown in Table III, incorporating the HFE module improves ESAD performance by 0.41%–6.40%, demonstrating the effectiveness of multi-level feature fusion in enhancing feature representation. This improvement arises because the HFE module integrates complementary information from different feature levels, thereby enabling the model to better capture both fine-grained details and high-level semantic contexts. In addition, Table III shows that the proposed HSFE module can significantly enhance ESAD performance. Specifically, the $F1$ increases by 14.57%–66.99% compared to the baseline without any additional modules. A similar trend is observed when HSFE is combined with HFE or SACC, with an $F1$ increase of 8.76%–67.28%. These results indicate that hyperbolic-based feature embedding is more effective than its Euclidean counterpart in representing complex and hierarchical data structures. Moreover, by leveraging the exponential growth property of hyperbolic geometry, HSFE can nonlinearly amplify differences between features and generate more discriminative feature representations, which is particularly beneficial for ESAD. In addition, the integration of the SACC module further improves the model's performance in ESAD, which surpasses the performances of both the baseline and other configurations, including the HFE-only, HSFE-only, and combined HFE+HSFE variants. Specifically, it yields $F1$ improvements of 20.35%–64.48% and 15.18%–64.05% compared with the baseline and HFE-only methods, respectively, and achieves a 7.09%–26.55% gain over the HSFE-only configuration. According to Table III, the ESAD performance is further improved by applying all three components together.

The results of the ablation experiment clearly demonstrate that each of the three proposed components contributes to improving ESAD accuracy, with the best performance achieved when all three components are integrated.

### B. Experimental Results and Analysis on the ESAD Dataset

To evaluate the effectiveness of the proposed method, a comprehensive comparison of the proposed H2PNet with popular anomaly detection methods was conducted on the ESAD dataset, and the quantitative results are reported in Table IV. The results demonstrate that the proposed H2PNet achieved the best performance, outperforming the classical and advanced anomaly detection and CD methods. Specifically, compared with the traditional anomaly detection method that is

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TABLE III
ABLATION STUDY RESULTS (%) OF THE PROPOSED MODULES APPLIED TO THE ESAD DATASET

| HFE | HSFE | SACC | Landslide | Forest fire | War | Green tide | Explosion | Volcano | Flood | Grassland fire |
|---|---|---|---|---|---|---|---|---|---|---|
| × | × | × | 3.99 | 19.22 | 22.82 | 2.17 | 49.55 | 26.58 | 20.99 | 43.52 |
| ✓ | × | × | 6.17 | 19.63 | 24.41 | 3.94 | 55.95 | 28.33 | 25.43 | 44.79 |
| × | ✓ | × | 70.98 | 53.37 | 59.24 | 64.19 | 64.12 | 66.96 | 56.35 | 73.50 |
| × | × | ✓ | 68.47 | 66.99 | 58.92 | 55.03 | 69.90 | 71.72 | 65.38 | 75.51 |
| ✓ | ✓ | × | 73.45 | 56.65 | 59.84 | 67.59 | 65.21 | 68.33 | 58.01 | 74.51 |
| ✓ | × | ✓ | 70.22 | 69.31 | 61.34 | 55.62 | 71.33 | 73.99 | 69.53 | 77.97 |
| × | ✓ | ✓ | 80.11 | 79.92 | 76.89 | 71.28 | 78.66 | 82.33 | 74.56 | 87.67 |
| ✓ | ✓ | ✓ | **81.87** | **82.87** | **78.66** | **75.62** | **81.95** | **87.39** | **75.49** | **91.56** |

TABLE IV
QUANTITATIVE RESULTS (%) OF THE COMPARED METHODS ON THE ESAD DATASET

| Event | Index | Traditional Methods | | DL-Based Methods | | | | H2PNet |
|---|---|---|---|---|---|---|---|---|
| | | RX | EBL | TAEF | DFAN | DWT-P-CSA | CAAE | |
| Landslide | Precision | 4.48 | 0.03 | 18.70 | 11.21 | 25.57 | 0.03 | 81.12 |
| | Recall | 85.15 | 79.18 | 58.89 | 80.77 | 71.09 | 92.97 | 82.63 |
| | $F1$ | 8.52 | 0.06 | 28.39 | 19.68 | 37.61 | 0.07 | **81.87** |
| Forest fire | Precision | 33.14 | 78.75 | 30.10 | 40.30 | 59.25 | 73.61 | 92.29 |
| | Recall | 51.45 | 75.19 | 70.97 | 76.24 | 89.62 | 87.96 | 75.18 |
| | $F1$ | 40.31 | 76.96 | 42.27 | 52.73 | 71.34 | 80.15 | **82.87** |
| War | Precision | 59.72 | 46.32 | 69.44 | 77.77 | 57.17 | 65.22 | 81.36 |
| | Recall | 43.75 | 28.75 | 40.79 | 38.47 | 35.05 | 64.48 | 76.14 |
| | $F1$ | 50.51 | 35.48 | 51.39 | 51.48 | 43.36 | 65.97 | **78.66** |
| Green tide | Precision | 14.35 | 61.24 | 32.42 | 24.99 | 70.23 | 78.12 | 72.77 |
| | Recall | 49.20 | 87.48 | 39.84 | 30.92 | 70.43 | 71.81 | 78.70 |
| | $F1$ | 22.22 | 72.05 | 35.75 | 27.64 | 70.33 | 74.83 | **75.62** |
| Explosion | Precision | 60.58 | 61.39 | 91.68 | 62.27 | 83.09 | 86.72 | 82.78 |
| | Recall | 72.64 | 84.49 | 71.59 | 71.28 | 78.30 | 78.72 | 81.13 |
| | $F1$ | 66.06 | 71.11 | 80.40 | 66.47 | 80.63 | **82.53** | 81.95 |
| Volcano | Precision | 29.23 | 28.04 | 16.51 | 30.89 | 42.19 | 88.81 | 88.27 |
| | Recall | 23.36 | 21.15 | 69.39 | 44.69 | 48.87 | 69.69 | 86.53 |
| | $F1$ | 25.97 | 24.11 | 26.68 | 36.53 | 45.29 | 78.10 | **87.39** |
| Flood | Precision | 61.40 | 79.81 | 74.74 | 62.76 | 60.04 | 69.96 | 81.42 |
| | Recall | 35.16 | 66.51 | 43.14 | 35.00 | 45.63 | 84.30 | 70.36 |
| | $F1$ | 44.71 | 72.56 | 54.71 | 44.94 | 51.85 | **76.46** | 75.49 |
| Grassland fire | Precision | 64.45 | 94.43 | 63.09 | 76.48 | 81.02 | 99.53 | 99.38 |
| | Recall | 2.85 | 72.45 | 5.54 | 9.21 | 53.87 | 81.04 | 84.88 |
| | $F1$ | 5.46 | 82.74 | 10.19 | 16.45 | 64.72 | 89.34 | **91.56** |

based on a single-temporal remote sensing image, the proposed H2PNet demonstrates a substantial accuracy advantage across all the scenarios in the ESAD dataset, with improvements in the $F1$ score, which ranges from 15.89% to 86.10%. The performance gap arises because the RX detector assumes multivariate Gaussian distributions for background characterization. This fundamental assumption becomes inadequate when real-world scenarios with heterogeneous backgrounds and target signatures that deviate from linear separability are considered. Moreover, the traditional anomaly detection method based on a single-temporal remote sensing image ignores the incorporation of historical prior knowledge, which often results in detection results that merely deviate from local or background statistics rather than representing semantically meaningful ESAs. Compared with the traditional anomaly detection method, the introduction of DL techniques markedly enhances both the accuracy and robustness of the anomaly detection algorithm based on a single-temporal remote sensing image. For example, on the ESAD dataset, the TAEF method achieves an improvement of 0.11%–19.87% in the $F1$ score compared with the RX algorithm. Similarly, relative to RX, DFAN and DWT-P-CDA yield $F1$ score improve-

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ment of 0.23%–12.42% and 7.14%–59.26%, respectively, on the ESAD dataset. These improvements arise primarily from the ability of DL models to automatically extract high-level spatial–spectral features from remote sensing images, capture complex nonlinear patterns, and effectively integrate contextual and semantic information. Despite these advantages, DL-based anomaly detection methods tend to underperform in scenarios where anomalies occupy a large proportion of the image, such as in the case of grassland fires. This is mainly due to their reliance on background modeling. When the proportion of anomalous regions is large, anomalous pixels participate in the estimation process of background parameters, thereby polluting the background model. Furthermore, these methods have yet to overcome the inherent deficiency of single-temporal anomaly detection methods; that is, they do not consider historical prior information. As a result, the detected anomaly results still reflect potential ESAs rather than confirmed anomalous events.

Compared with anomaly detection methods (RX, TAEF, DFAN, and DWT-P-CDA) that are based on single-temporal remote sensing image, CD-based methods tend to demonstrate superior accuracy and robustness. By incorporating temporal information, CD-based methods can explicitly capture the evolution of Earth surface features over time, thereby effectively suppressing temporally invariant background noise and enhancing sensitivity to real anomalous events. In particular, the introduction of the DL technique significantly improves the performance of anomaly detection. Specifically, the CAAE method yields an improvement of 0.01%–50.25% in the $F1$ score over the EBL method on the ESAD dataset. Furthermore, compared with single-temporal remote sensing image-based anomaly detection methods, it achieved a marked accuracy gain of 1.90%–83.88%. However, CD-based methods depend heavily on the availability and quality of accurately registered historical data. In this study, the proposed H2PNet can learn the historical steady-state prior of the normal Earth surface by incorporating spatiotemporal–spectral information during the training stage and then encoding it in a prototype set. During inference, H2PNet can perform anomaly detection using only single-temporal remote sensing image by computing the minimum hyperbolic distance between the feature representation of test samples and the prototypes in the prototype set via a dictionary lookup mechanism. As shown in Table IV, the proposed method achieves superior ESAD performance across various ESA types, thus demonstrating its effectiveness and robustness.

To intuitively demonstrate the advantages of the proposed method in ESAD, Figs. 7 and 8 present visual comparison results of various methods and locally enlarged views of partial results, respectively. As shown in Fig. 7, single-temporal remote sensing image-based methods tend to misidentify normal objects that deviate from the local pattern or background distribution as anomalies because of the absence of historical prior constraints. Moreover, when anomalies occupy a large portion of the image, these methods often suffer from missed detections, as the anomalous signals contaminate the background model. In contrast, CD-based methods generally achieve better performance when the historical reference data are of high quality. However, their effectiveness degrades significantly when the quality of the reference image decreases, such as when it is affected by cloud contamination, as demonstrated in the landslide case shown in Fig. 7. Compared with the previous methods, the method proposed in this study can accurately detect ESAs that deviate from learned steady-state prior by modeling historical stable states and enhancing feature space separation between anomalies and stable states.

## V. Discussion

### A. Effectiveness Analysis of H2PNet in ESAD

The results of ablation studies and comparative experiments have demonstrated the effectiveness of the proposed method in ESAD tasks. To gain an intuitive understanding of model performance and to illustrate its advantages in ESAD, the ASs generated during model inference are visualized, as shown in Fig. 9. The results reveal that the proposed method consistently detected anomalous regions with high ASs, even in complex scenes, against varying backgrounds, and in low-contrast environments, while maintaining low scores for the normal Earth surface, thus indicating strong discrimination between the foreground and background. This capability is attributed primarily to three factors. First, the enhanced hyperbolic space can represent high-dimensional complex data in low-dimensional spaces by capturing hierarchical structures in images, thereby extracting richer feature representations from remote sensing images and constructing a robust steady-state prior. Second, the proposed SACC module further enhances the representativeness of the constructed prior by reinforcing intraclass consistency among normal samples. Finally, unlike Euclidean space, the exponentially expanding geometric property of hyperbolic space enables the model to further amplify the distinction between normal and anomalous samples, thereby facilitating clearer separation of anomalies from complex backgrounds. Notably, the regions outlined by red dashed circles in Fig. 9(a) and (e) indicate that the model has fine-grained perception capabilities. Even when the proportion of anomalous areas is relatively small, the model can still overcome the interference from complex backgrounds and effectively distinguish between normal and anomalous Earth surface. These results provide additional evidence of the robustness and effectiveness of the proposed method.

In addition, experimental results on the four cases with spatial nonoverlap characteristics (landslide, forest fire, war, and green tide) demonstrate that the model achieves effective ESAD even when the prior construction region does not completely overlap with the detection region. In the landslide and forest fire cases, despite the steady-state prior being constructed from only local subregions within the detection area, H2PNet rarely misidentifies areas outside these subregions as anomalies. In the war and green tide cases, H2PNet achieves cross-regional generalized detection based on the steady-state prior constructed from adjacent areas. This spatial generalization capability is mainly attributed to two factors: first, adjacent or local regions share continuity and similarity in geographical elements (e.g., terrain and LC); second, DL-based models can extract essential features and latent patterns

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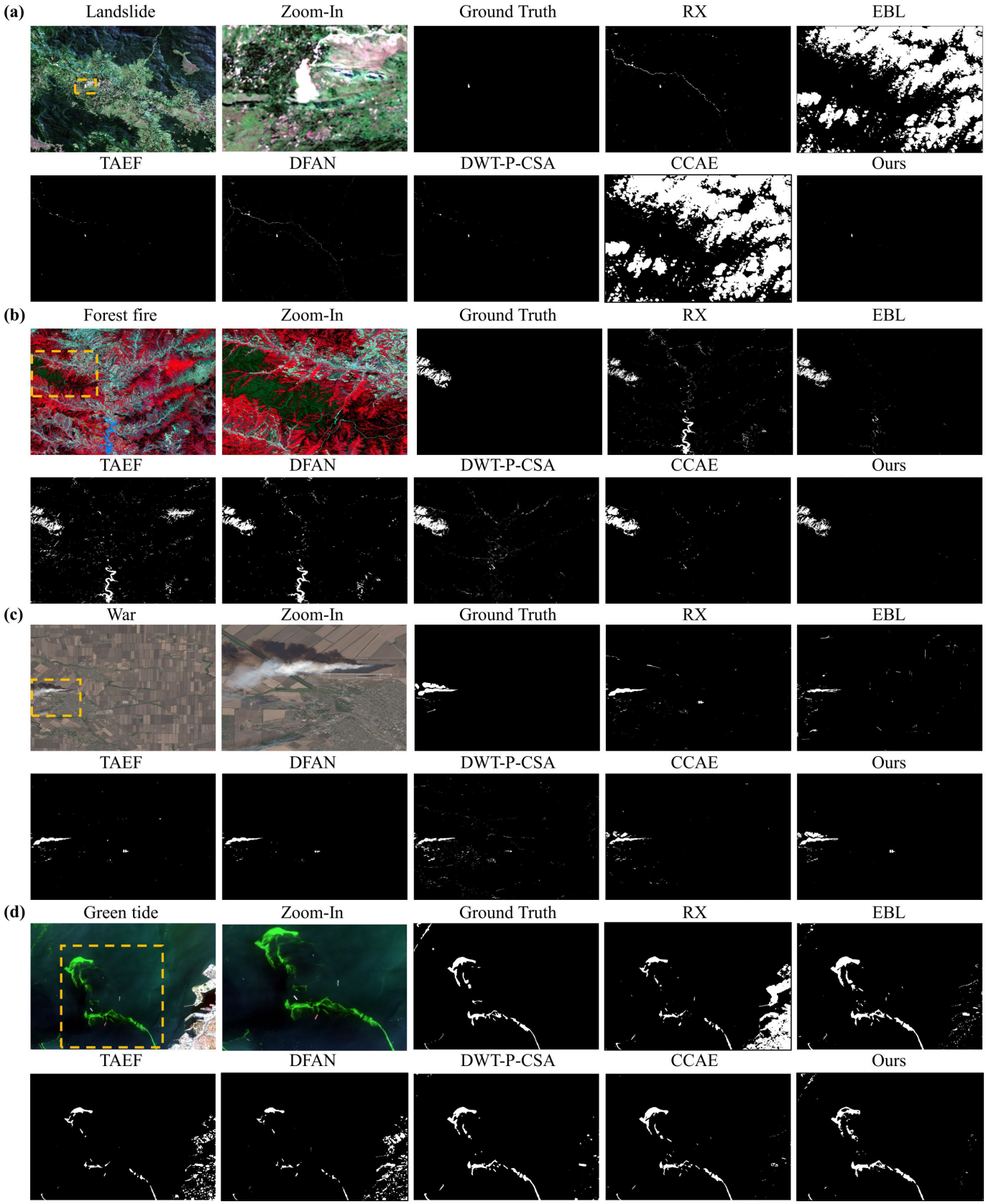



Fig. 7. ESAD results for (a) landslide, (b) forest fire, (c) war, (d) green tide, (e) explosion, (f) volcano, (g) flood, and (h) grassland fire data for H2PNet and comparison methods.

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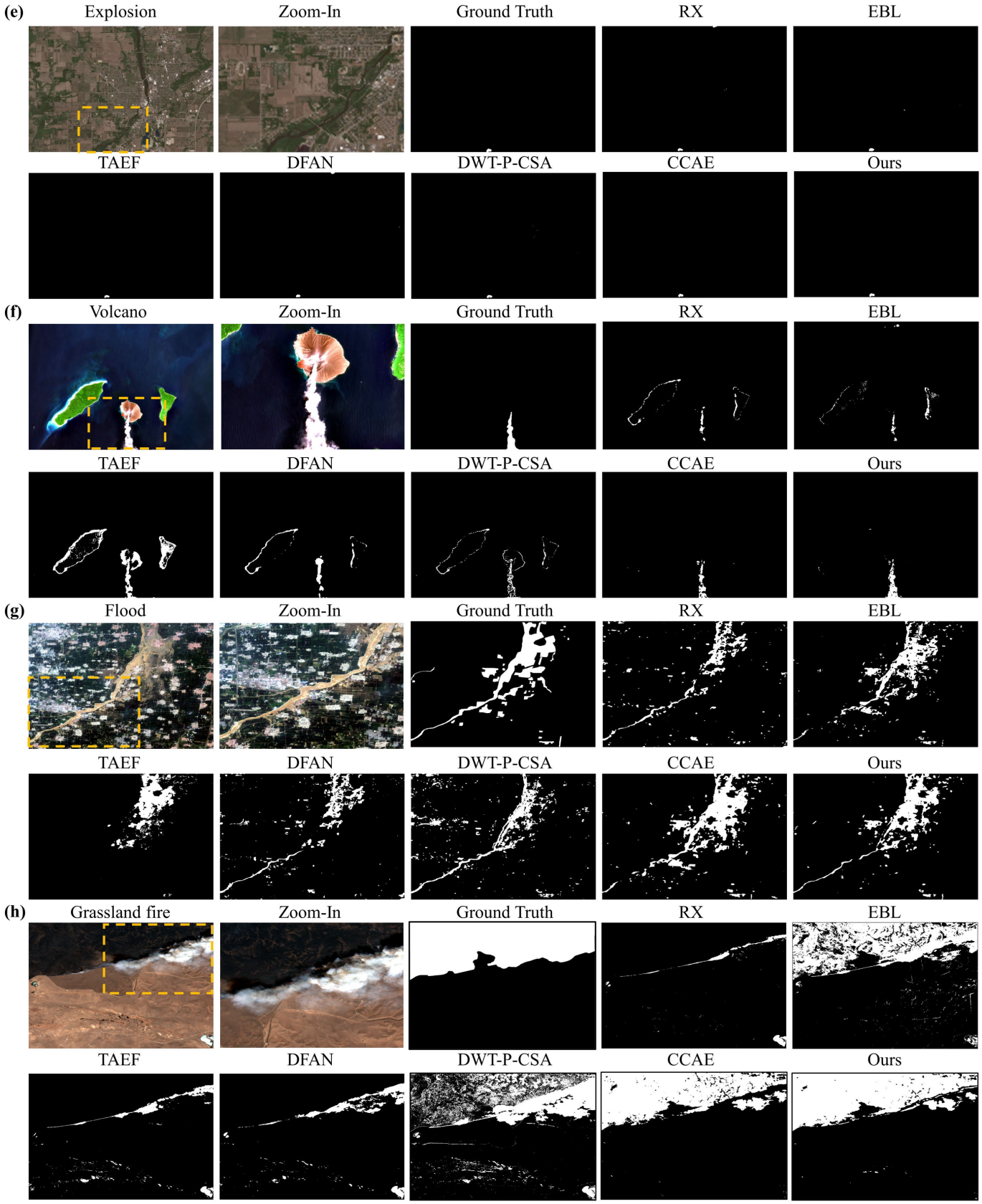



Fig. 7. *(Continued.)*ESAD results for (a) landslide, (b) forest fire, (c) war, (d) green tide, (e) explosion, (f) volcano, (g) flood, and (h) grassland fire data for H2PNet and comparison methods.

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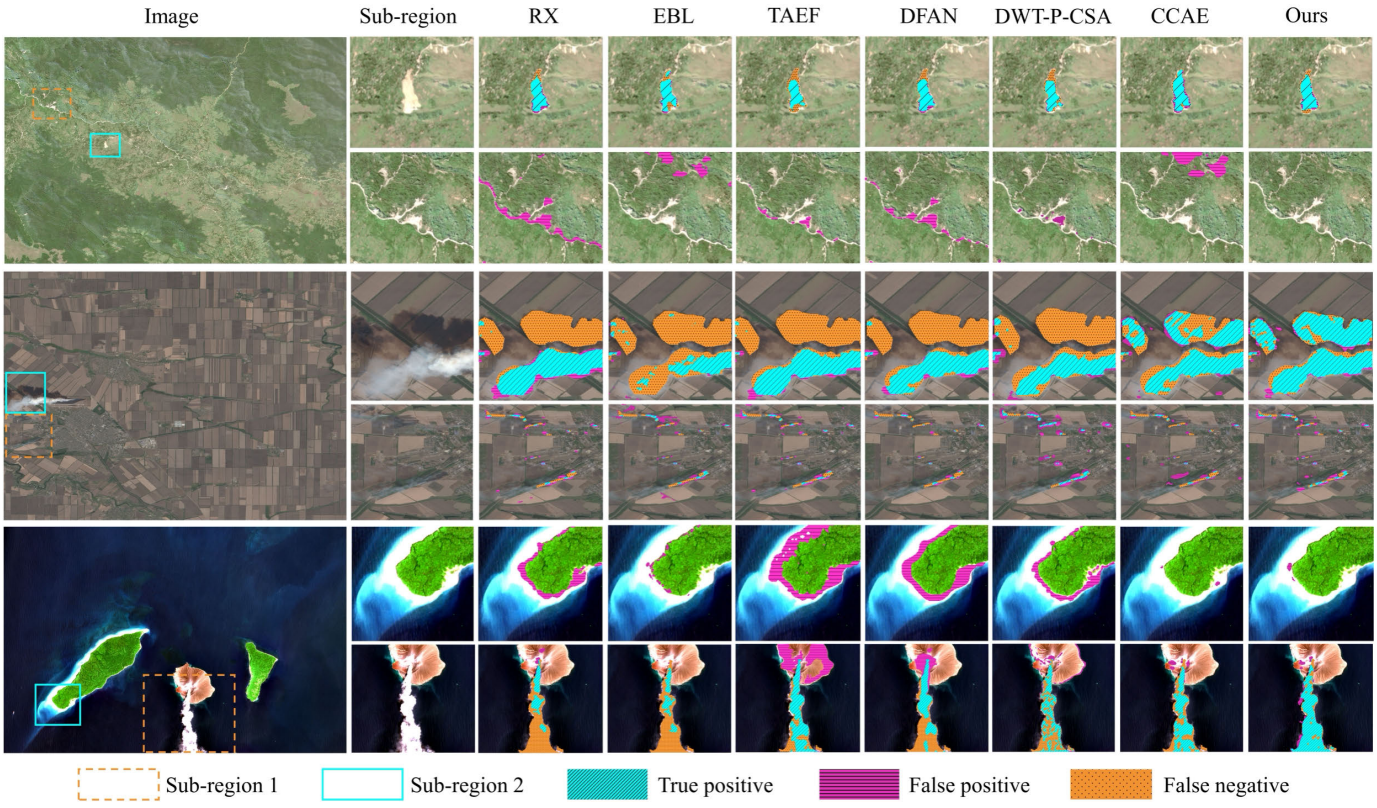



Fig. 8. Local detail views of partial results. Positive and negative denote anomalous and normal regions, respectively. More false positive pixels indicate regions incorrectly predicted as anomalous, and more false negative pixels indicate regions of true anomalies that were missed.

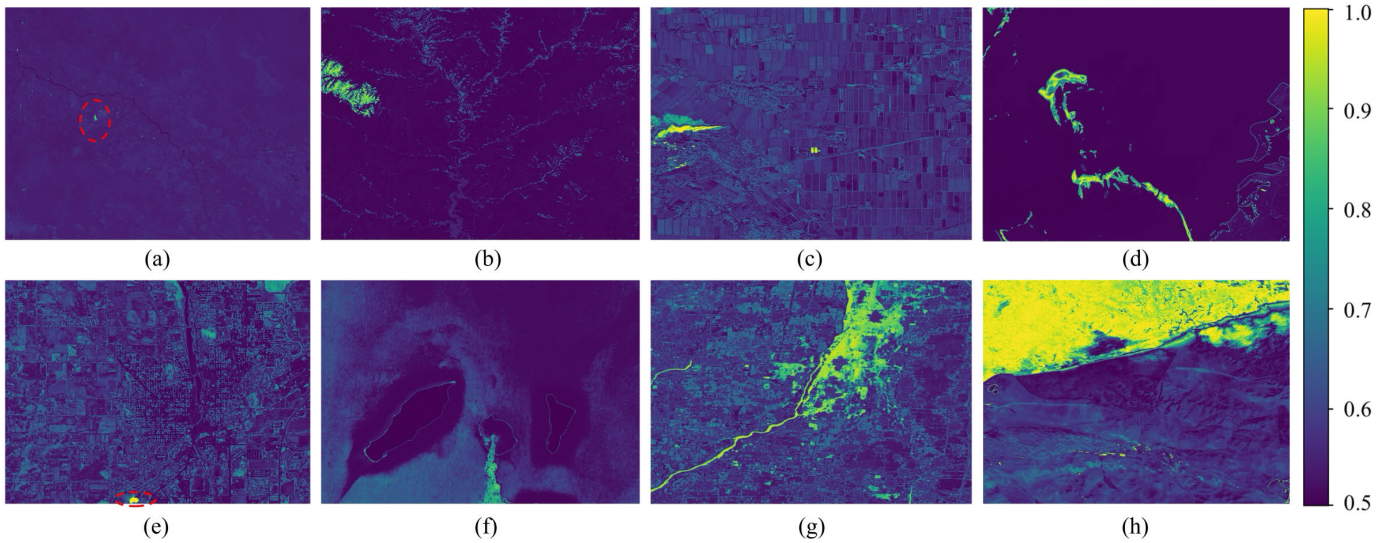



Fig. 9. Visualization results for the ASs of the model on the ESAD dataset. (a) Landslide. (b) Forest fire. (c) War. (d) Green tide. (e) Explosion. (f) Volcano eruption. (g) Flood. (h) Grassland fire.

of geographical phenomena from training samples. Through multilevel feature fusion and spatial context modeling, H2PNet effectively recognized similar Earth surface patterns across different regions, thereby achieving robust spatial generalization. These characteristics endowed H2PNet with significant practical value in real-world application scenarios where historical observations are spatially incomplete due to cloud cover.

### *B. Computational Cost and Prior Volume*

To evaluate the computational efficiency of the proposed method, particularly its inference speed on single-temporal remote sensing image, inference speeds were measured for H2PNet and benchmark methods on the ESAD dataset, as shown in Table V. Compared with DL-based methods, traditional methods such as RX and EBL achieve faster anomaly

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TABLE V
COMPUTATIONAL COSTS AND PRIOR VOLUMES OF VARIOUS METHODS ON THE ESAD DATASET

| Method | Index | Landslide | Forest fire | War | Green tide | Explosion | Volcano | Flood | Grassland fire |
|---|---|---|---|---|---|---|---|---|---|
| RX | Time (S) | 4.19 | 19.54 | 8.59 | 0.61 | 1.66 | 2.20 | 0.83 | 5.41 |
| | Volume (M) | - | - | - | - | - | - | - | - |
| EBL | Time (S) | 1.56 | 8.46 | 3.23 | 0.23 | 0.67 | 0.68 | 0.64 | 2.09 |
| | Volume (M) | 206.68 | 816.63 | 424.47 | 30.46 | 75.73 | 96.57 | 31.62 | 300.41 |
| TAEF | Time (S) | 91.21 | 511.37 | 191.20 | 13.00 | 32.13 | 44.07 | 10.68 | 219.93 |
| | Volume (M) | 0.07 | | | | | | | |
| DFAN | Time (S) | 57.04 | 240.31 | 99.61 | 11.47 | 27.33 | 28.28 | 8.12 | 88.92 |
| | Volume (M) | 0.04 | | | | | | | |
| DWT-P-CSA | Time (S) | >300 | >300 | >300 | 78.31 | 169.24 | 221.85 | 60.43 | >300 |
| | Volume (M) | 0.58 | | | | | | | |
| CAAE | Time (S) | 13.97 | 51.43 | 22.67 | 2.15 | 4.86 | 6.11 | 1.55 | 15.23 |
| | Volume (M) | 206.68 | 816.63 | 424.47 | 30.46 | 75.73 | 96.57 | 31.62 | 300.41 |
| H2PNet | Time (S) | 16.63 | 48.47 | 28.53 | 7.69 | 9.00 | 11.06 | 5.04 | 18.86 |
| | Volume (M) | 0.30 | | | | | | | |

inference speeds. This results from their reliance on simple mathematical operations, eliminating the need for forward propagation through deep neural networks. In particular, the CD-based EBL method records the fastest inference speed by directly computing the differences between preevent and postevent images. Although the introduction of DL-based techniques reduces the inference speed to a certain extent, the proposed H2PNet remains highly competitive among DL-based methods in terms of inference efficiency. It is worth noting that CD-based methods require additional time for data registration and transmission, which is not included in the statistics reported in Table V. In addition, the prior volume for various ESAD methods is reported in this study. The RX method, which performs image operations directly, does not require any prior information. DL-based methods that use single-temporal remote sensing image need to store only lightweight model state files.

Moreover, as mentioned previously, the hierarchical perception capability of hyperbolic space enables the proposed H2PNet to represent more complex data structures in lower dimensions, which reduces the requirement for complex models in high-precision ESAD, thereby making the stored prior more lightweight. In contrast, CD-based methods require registered historical imagery, which not only increases the complexity of data preparation but also results in greater storage overhead. Notably, as shown in Table V, only the model's inference speed and the volume of historical steady-state prior knowledge required during inference are reported in this study, as prior knowledge is constructed primarily on the ground, with only the inference module designed for on-orbit deployment and application.

Overall, the proposed method shows competitive performance in terms of ESA inference speed, detection accuracy, and storage requirements, thus demonstrating strong potential for on-orbit deployment and application in resource-constrained environments.

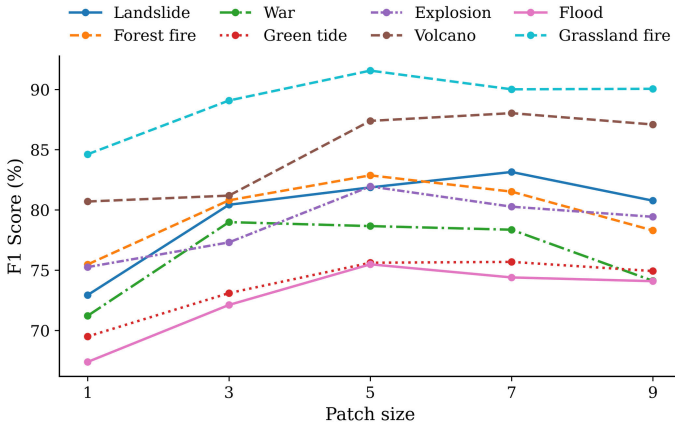



Fig. 10. Effect of different patch sizes on ESAD.

### C. Sensitivity Analysis of Key Parameters

In the proposed H2PNet model, the patch size, curvature $c$, and adjustment coefficient $\beta$ will potentially influence the performance of ESAD. To evaluate the effects of these parameters, additional experiments were conducted to analyze the parametric sensitivity of the H2PNet.

*1) Patch Size:* In the ESAD task, the patch size directly influences the model's ability to extract spectral–spatial features. To evaluate the effect of patch size on ESAD, accuracy comparisons were conducted across five different patch sizes (1×1, 3×3, 5×5, 7×7, and 9×9) on the ESAD dataset, with results shown in Fig. 10. Overall, patch-based methods generally achieve superior detection performance over pixel-based approaches in ESA identification. This improvement stemmed from the effective utilization of spatial relationships between adjacent pixels, which provides richer contextual information. Such context enables the H2PNet to better capture the structure and semantic characteristics of the imagery, thereby facili-

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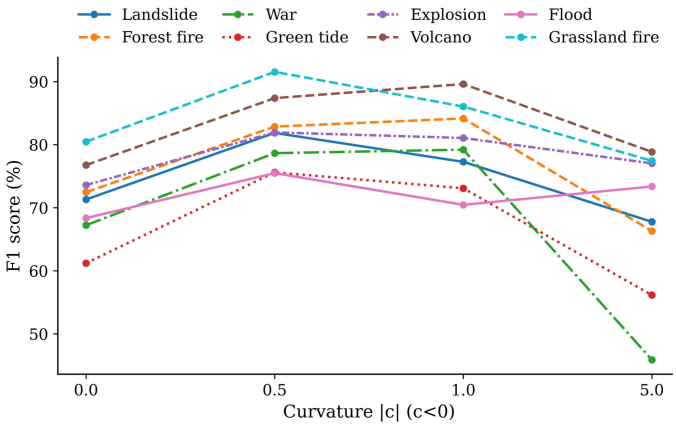



Fig. 11. Sensitivity analysis of curvature parameter in ESAD.

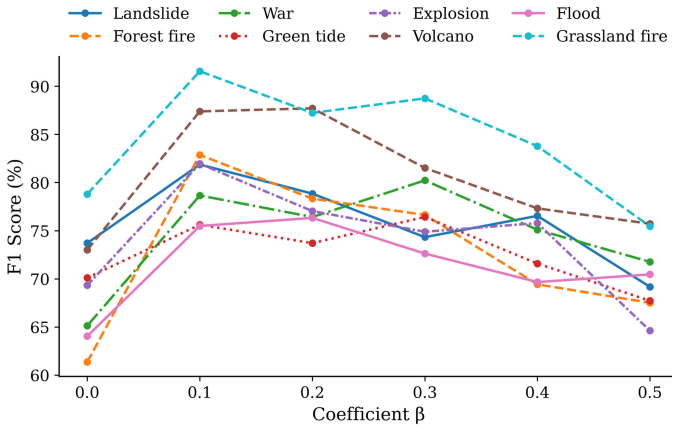



Fig. 12. Sensitivity analysis of the adjustment coefficient in ESAD.

tating a more precise representation of normal Earth surface patterns. Specifically, as patch size increases from $1 \times 1$ to $5 \times 5$, model performance consistently improved across all anomaly scenarios, demonstrating the benefit of incorporating spatial contextual information. For instance, volcano eruption detection accuracy rises from 80.70% ($1\times1$) to 87.39% ($5\times5$), while grassland fire increases from 84.62% to 91.56%. When the patch size reaches $7\times7$, detection performance remains stable compared to $5\times5$ across all anomaly categories, with differences within a narrow range. However, further expanding to $9 \times 9$ results in performance degradation for most anomaly scenarios, which may be explained by a larger patch size introducing redundant or irrelevant background information that impairs detection accuracy. Given that $7 \times 7$ patches incur higher computational costs without commensurate performance gains, $5 \times 5$ was adopted as the optimal configuration, achieving the best balance between detection performance and computational efficiency.

*2) Curvature $c$* : In hyperbolic space, the curvature directly affects the space structure and subsequently influences the model performance. Therefore, additional experiments were conducted to investigate how ESAD performance varies with curvature $c$, with results illustrated in Fig. 11. When curvature $c$ is set to zero, the model degenerates to a conventional Euclidean representation with limited expressive capacity, where high-dimensional feature distributions are susceptible to concentration and overlap, thereby blurring interclass boundaries and limiting anomaly discrimination capability. As curvature gradually decreases from zero (i.e., increasing negative curvature magnitude), transitioning the embedding space into hyperbolic geometry, model performance improves within a specific curvature range. This improvement suggests that hyperbolic space can effectively improve ESAD capability through enhanced feature embedding representation and nonlinear stretching of feature differences. However, when the absolute curvature becomes excessively large (e.g., $|c| = 5$), the model performance begins to decline. This degradation is primarily attributed to numerical instability in feature mapping, distorted embeddings in extreme curvature regions, and gradient amplification during optimization. These factors collectively compromise local structure preservation and impair ESAD capability.

*3) Adjustment Coefficient $\beta$* : The adjustment coefficient $\beta$ regulates the relative contribution of contrastive clustering loss $\mathcal{L}_{\mathcal{H}c}$ to classification loss $\mathcal{L}_{\text{cla}}$, where an appropriately chosen $\beta$ can effectively enhance the representativeness and reliability of the steady-state prior. The effect of the adjustment coefficient $\beta$ on model performance is shown in Fig. 12. When $\beta = 0$, the contrastive clustering loss does not participate in model optimization, and the semantic enhanced contrastive clustering module loses its ability to constrain the feature distribution. As $\beta$ increases to the range of [0.1, 0.3], H2PNet exhibits stable anomaly detection performance with $F1$ scores varying by less than 3% (relative standard deviation), demonstrating robustness to this adjustment coefficient. However, further increasing $\beta$ imposes excessively strong contrastive clustering constraints, which overcompress intraclass features and suppresses the natural variability of normal Earth surface patterns. Consequently, the obtained prototype set will fail to fully represent the natural variability of historical steady-state distributions, making the model overly sensitive to subtle variations in normal samples and leading to a decline in ESAD performance. Based on the stable performance observed in [0.1, 0.3], $\beta$ was set to 0.1 in this study.

### D. Influence of Historical Data Quantity and Time Interval

In this study, historical data are the foundation for constructing the steady-state prior, directly determining the performance of ESAD. The quality of the steady-state prior is critically influenced by two factors: the quantity of historical data and the time interval between data points. Therefore, additional experiments were conducted to discuss the impact of data volume and time intervals on model performance. Specifically, this study selected four cases from the ESAD dataset, namely, landslides, war, green tide, and explosion, and constructed a steady-state prior using historical remote sensing images that reflect normal Earth surface conditions, as described in Table II. Then, the effect of varying historical image quantities on model stability and ESAD was evaluated, with the experimental results shown in Table VI. The results in Table VI indicate that the model performs consistently stable

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TABLE VI
EFFECT OF HISTORICAL IMAGE QUANTITY ON ESAD

| Event | Record | Number of Historical Images | | |
|---|---|---|---|---|
| | | 1 | 2 | 3 |
| Landslide | Date | 2024-05-23 | 2024-05-23, 2024-04-23 | 2024-05-23, 2024-04-23, 2024-04-08 |
| | $F1$ | 79.83 | 78.92 | 81.87 |
| War | Date | 2024-09-16 | 2024-09-16, 2024-09-11 | 2024-09-16, 2024-09-11, 2024-09-08 |
| | $F1$ | 72.84 | 76.53 | 78.66 |
| Green tide | Date | 2019-07-23 | 2019-07-23, 2019-07-18 | 2019-07-23, 2019-07-18, 2019-07-11 |
| | $F1$ | 75.26 | 74.52 | 75.62 |
| Explosion | Date | 2021-06-04 | 2021-06-04, 2021-05-10 | 2021-06-04, 2021-05-10, 2021-04-30 |
| | $F1$ | 79.36 | 80.82 | 81.95 |

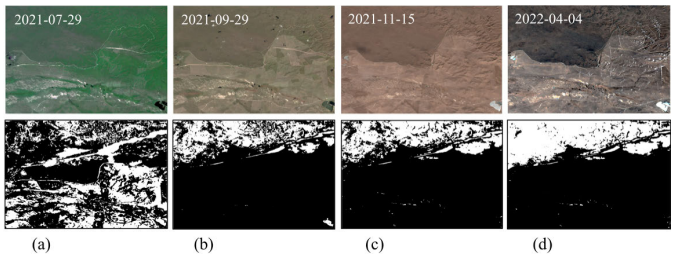



Fig. 13. Effect of phenological variability on ESAD performance. (a) $F1$ = 30.96%. (b) $F1$ = 66.96%. (c) $F1$ = 74.07%. (d) $F1$ = 89.04%.

performance under varying quantities of historical remote sensing images, with only marginal improvements observed when increasing from one to three images. This is primarily because the frequency of ESAs is low, and thus, the Earth surface remains relatively stable over a certain period before the anomaly occurs, allowing even a single image to reflect the steady-state characteristics of the Earth surface to a certain extent. However, a single image cannot adequately capture relatively low-frequency yet normal Earth surface features such as precipitation. Therefore, integrating multiple historical images can mitigate the potential impact of such short term but normal variations on the model to some extent.

Furthermore, for the time interval of historical images, multiple historical remote sensing images that were available prior to the anomaly and temporally close to its occurrence (i.e., with minimal cloud interference and phenological differences) were selected as the basis for constructing the steady-state prior, rather than relying on historical images with a long temporal gap. This is because phenological changes caused by seasons may alter the feature distribution of Earth surface objects, especially in high-latitude regions. If historical images with a large time span and significant phenological differences are used for modeling the steady-state prior of the Earth surface, it will become challenging for the constructed steady-state prior to distinguish between seasonal variations and true ESAs. To evaluate the impact of steady-state prior constructed under different phenological conditions on ESAD, an experiment was conducted using the grassland fire in Mongolia on April 19, 2022, as a case study, with results shown in Fig. 13. As can be seen from the figure, when the steady-state prior is constructed from image with phenological conditions most similar to the anomaly event, the model achieves optimal ESAD performance, as shown in Fig. 13(d). Conversely, when phenological features differ significantly, the model tends to confuse seasonal changes with true anomalies, as shown in Fig. 13(a).

In conclusion, considering the impacts of precipitation, phenological changes, and other factors, this study adopts three available remote sensing images (with minimal cloud interference and minimal phenological differences) within two months before the anomaly occurrence as the basis for steady-state prior modeling. Furthermore, given that ESAs cannot be predicted in advance during real-time satellite observations, this study will further explore methods for constructing the steady-state prior that cover the entire year temporally and can reflect phenological changes in future research. For instance, steady-state prior intervals may be divided by seasons, months, or satellite revisit cycles to develop a segmented paradigm for Earth surface steady-state prior modeling.

### *E. Pros and Cons of H2PNet*

In this section, the application value and shortcomings of the proposed H2PNet are explored. To meet the demand for on-orbit ESAD, H2PNet is proposed, which is a lightweight ESAD algorithm designed for single-temporal remote sensing imagery that achieves rapid, reliable, and general ESAD. In contrast to existing single-temporal methods, H2PNet strikes a balance between generality and accuracy. First, H2PNet eliminates the reliance on predefined anomaly categories in direct ESAD methods by modeling the historical steady states of the normal Earth surface, thereby enabling general ESAD. Second, the integration of spatiotemporal–spectral historical prior addresses the limitations of background modeling methods that are based solely on spectral and spatial features, which often result in the detection of only background-deviating "potential ESAs" rather than semantically meaningful ESA events. More importantly, the proposed H2PNet can nonlinearly amplify feature differences between normal and anomalous objects, thereby mitigating the effects of ambiguous decision boundaries on the accuracy of ESAD. The results of comparative experiments with CD-based methods further demonstrate the advantages of H2PNet in terms of accuracy, storage, and

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computational efficiency, thus offering stronger support for its potential in on-orbit deployment and application.

The proposed method is designed primarily to meet the demand for on-orbit immediate ESAD, where the model relies on remotely sensed data acquired in real time to ensure that anomaly signals are fully preserved. For example, when a landslide has just occurred, its spectral and spatial characteristics differ significantly from those of normal surfaces. However, over time, its spectral features may gradually approach or even become similar to those of bare soil, which inevitably increases the difficulty of anomaly detection and leads to a decline in the model's accuracy in recognizing anomalies at later stages.

## VI. Conclusion

A novel ESAD model named H2PNet for on-orbit deployment and applications, which integrates historical steady-state prior construction of the normal Earth surface with anomalous feature separation to enable accurate ESAD from single-temporal remote sensing imagery, was proposed in this study. The designed enhanced hyperbolic space network employs its hierarchical perception capability to extract richer features from remote sensing data, thereby significantly improving the model's capacity to characterize the normal Earth surface. SACC is then performed to integrate the semantic context and contrastive clustering, which enhances the representativeness of the constructed prior. During inference, the proposed hyperbolic distance-based scoring function obtains satisfactory ESAD by leveraging the nonlinear amplification of feature differences in hyperbolic space. The results of extensive experiments demonstrate the efficiency and robustness of H2PNet in unified ESAD, which highlights its strong potential for on-orbit deployment and application.

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